

Introduction to Probability

Mingxun Zhou (mingxunz@ust.hk)

COMP2711 Discrete Mathematical Tools

1 Probability models

In Powerball, a play selects five distinct white numbers from 1–69 (order does not matter) and one red number from 1–26. Matching all six wins the jackpot. Someone claims its probability is $1/2$ because *either you hit it or you do not*. Two possible descriptions do not make the two events equally likely. We must specify the possible outcomes and how they are selected. There are $\binom{69}{5} \cdot 26 = 292,201,338$ equally likely results, and exactly one matches a given play. The jackpot probability is therefore $1/292,201,338$, consistent with the [official Powerball odds](#).

MUST UNDERSTAND: Sample space and event

An **experiment** is a procedure that produces an outcome. An individual **outcome** ω is one possible result. Its **sample space** S is the set of all possible outcomes. An **event** E is a subset of S : the outcomes satisfying a specified condition. The event occurs when the observed outcome satisfies $\omega \in E$.

For example, draw one ticket from 100 tickets labelled $1, \dots, 100$. An outcome is one ticket number. For the event that the number is odd,

$$S = \{1, 2, \dots, 100\}, \quad E = \{1, 3, 5, \dots, 99\}.$$

For example, $7 \in E$, whereas $42 \notin E$. An outcome is an element; an event is a set of outcomes.

MUST UNDERSTAND: Classical probability model

The classical model assumes that S is finite and nonempty and that *every outcome in S is equally likely*. Then, for every event $E \subseteq S$,

$$\Pr[E] = \frac{|E|}{|S|}.$$

The numerator counts outcomes satisfying the event; the denominator counts all outcomes. Use the same representation in both counts. Since $0 \leq |E| \leq |S|$,

$$0 \leq \Pr[E] \leq 1, \quad \Pr[\emptyset] = 0, \quad \Pr[S] = 1.$$

For a simplified lottery, only ticket 1 of our 100 tickets wins. If every ticket is equally likely, then

$$\Pr[\text{jackpot}] = \frac{1}{100}, \quad \Pr[\text{no jackpot}] = \frac{99}{100}.$$



The error is treating two events as equally likely simply because there are two of them. Equal likelihood must come from the experiment, not from its labels.

General discrete probability model

For an individual outcome $x \in S$, write $\Pr[x]$ as shorthand for $\Pr[\{x\}]$. In the classical model, $\Pr[x] = 1/|S|$ for every x .

More generally, S can be finite or even countably infinite. Assign a probability to each outcome, subject to

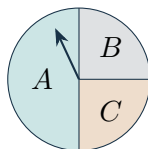
$$\Pr[x] \geq 0 \quad (x \in S), \quad \sum_{x \in S} \Pr[x] = 1.$$

Different outcomes may have different probabilities. For any event $E \subseteq S$,

$$\Pr[E] = \sum_{x \in E} \Pr[x].$$

That is, add the probabilities of all outcomes in the event. For a finite uniform space, this becomes $\sum_{x \in E} 1/|S| = |E|/|S|$.

Example: An unequal spinner. A spinner lands uniformly by angle. Half the circle is labelled A , one quarter B , and one quarter C . Recording the label gives $S = \{A, B, C\}$, with $\Pr[A] = 1/2$ and $\Pr[B] = \Pr[C] = 1/4$. For $E = \{A, B\}$, we have $\Pr[E] = 1/2 + 1/4 = 3/4$, not $2/3$.



A countably infinite example is $S = \{1, 2, 3, \dots\}$ with $\Pr[k] = 2^{-k}$: the geometric series sums to 1. A countably infinite sample space cannot give every outcome the same probability: a common value of zero sums to zero, while a positive common value gives an unbounded sum. The three models below return to finite, equally likely outcomes.

2 Model 1: Independent random trials

In independent trials, knowing earlier results does not change the chances of later results. For our uniform model, each of t trials has the same m equally likely possible results. One outcome records all t results in order:

$$S = \{1, \dots, m\}^t, \quad |S| = m^t.$$

Independence and uniform choices together make these sequences equally likely. Independence alone does not imply uniformity: independent biased coin flips do not give equally likely strings.

Example: Exactly two heads

Flip a fair coin three times independently. What is the probability of exactly two heads? One outcome is a three-letter H/T sequence. There are $2^3 = 8$ equally likely sequences. The event is

$$E = \{HHT, HTH, THH\}, \quad \Pr[E] = \frac{3}{8} = \frac{\binom{3}{2}}{2^3}.$$

Choosing the two head positions determines the complete sequence. The fair-coin assumption makes the choices equally likely; independence means earlier flips do not change the chances on a later flip. Both assumptions matter.

A reduction need not be uniform. Flip a fair coin twice, with the first result not affecting the second. The ordered strings HH, HT, TH, TT are equally likely. If we record only the number of heads, the three possibilities represent different numbers of strings:

Head count	0	1	2
Full outcomes	TT	HT, TH	HH
Probability	1/4	1/2	1/4

There are three possible head counts, but they do not each have probability 1/3.

Example: Drawing with replacement

A bag contains ten distinct balls labelled $1, \dots, 10$. Draw uniformly, return the ball, and mix before each new draw. Assume the draws are independent. What is the probability of drawing 3, 7, 1 in that order? One outcome is an ordered triple; repeated labels are allowed. There are 10^3 equally likely triples, and just one is $(3, 7, 1)$, so

$$\Pr[\text{draw } 3, 7, 1 \text{ in order}] = \frac{1}{10^3} = \frac{1}{1000}.$$

3 Model 2: Random permutation

Shuffle a standard deck of 52 distinct cards, without jokers: 13 ranks and four suits, with one card of each rank in each suit. One outcome is the complete order of all 52 cards. A uniform shuffle makes all $52!$ orders equally likely.

In general, a uniform random permutation orders all N distinct objects, each exactly once. Its sample space consists of all $N!$ equally likely orders. Unlike the independent-trial model, selecting an object removes it from the choices for later positions.

Example: One specific shuffle

Shuffle the deck uniformly. What is the probability of one complete card order specified in advance? One outcome is an ordered list of all 52 cards. By the product rule, there are $52 \cdot 51 \cdots 1 = 52!$ such orders. The event containing just the prescribed order ω_0 has size 1, so

$$\Pr[\{\omega_0\}] = \frac{1}{52!}.$$

Here $52!$ counts outcomes; its reciprocal is the probability of a specific outcome.

Example: The first ace

Shuffle a standard 52-card deck uniformly and reveal cards until an ace appears. What is the probability that the first ace is $A\heartsuit$?

Solution 1: Direct counting. Let E be the set of complete deck orders whose first ace is $A\heartsuit$. There are $52!$ total orders, so $\Pr[E] = |E|/52!$. Choose the four ace positions in $\binom{52}{4}$ ways. Put $A\heartsuit$ in the earliest chosen position, arrange the remaining three aces in $3!$ ways, and arrange the 48 non-aces in the other positions in $48!$ ways. Thus

$$|E| = \binom{52}{4} 3! 48!, \quad \Pr[E] = \frac{\binom{52}{4} 3! 48!}{52!} = \frac{1}{4}.$$

Solution 2: Relative counting. Ignore the non-aces and ask which ace comes first. In a uniform shuffle, the four aces are symmetric, so each is equally likely to come first. Their probabilities sum to 1; therefore

$$\Pr[\text{first ace is } A\heartsuit] = \frac{1}{4}.$$

Formally, exchanging the labels of any two aces pairs the shuffles in which one comes first with those in which the other comes first.

More generally, fix any subset of k cards before observing a uniform shuffle. After discarding the other cards, all $k!$ relative orders of the selected cards are equally likely. Relabelling the selected cards pairs the full shuffles giving any one relative order with those giving any other, and uniformity preserves their probabilities. For the four aces, each ace is first in $3!$ of the $4!$ orders. This shortcut is justified by the uniform shuffle; it is not guaranteed for an arbitrary nonuniform shuffle.

Example: The first ace before the first king

Shuffle the 52-card deck uniformly and reveal cards one by one. What is the probability that the first ace appears before the first king? Keep only the four aces and four kings. Their relative order is uniform, so each of the eight cards is equally likely to come first. The event occurs exactly when that first card is one of the four aces. Thus

$$\Pr[\text{first ace before first king}] = \frac{4}{8} = \frac{1}{2}.$$

Example: All aces before the first king

In a uniform 52-card shuffle, what is the probability that all four aces appear before the first king? The event depends only on the relative order of the four aces and four kings. By the uniform-subset property, their $8!$ relative orders are equally likely.

The qualifying pattern is four aces followed by four kings, ignoring the other cards:



There are $4!$ ways to order the aces and $4!$ ways to order the kings. Consequently,

$$\Pr[\text{all aces before the first king}] = \frac{4!4!}{8!} = \frac{1}{\binom{8}{4}} = \frac{1}{70}.$$

Unlike the previous event, it is not enough for just one ace to precede the first king.

4 Model 3: Random drawing without replacement

Choose r of N distinct objects uniformly without replacement, with $1 \leq r \leq N$, and record only which objects were selected. A selected object cannot be chosen again, and draw order is ignored. For example, choosing three labels from $\{A, B, C, D, E\}$ might produce $\{A, C, E\}$. The orders A, C, E and E, A, C describe the same selection. One outcome is an r -element subset, so

$$|S| = \binom{N}{r}.$$

The subsets are equally likely because each has exactly $r!$ equally likely draw orders. This is the natural representation for a card hand.

Example: One specific five-card hand

Draw five cards uniformly without replacement and ignore their order. What is the probability of one particular hand H_0 specified in advance? One outcome is a five-card subset of the deck. There are $\binom{52}{5}$ hands. Each hand has $5!$ equally likely draw orders, so all hands are equally likely. Thus

$$\Pr[\{H_0\}] = \frac{1}{\binom{52}{5}} = \frac{1}{2,598,960}.$$

Example: Four of a kind

Draw five cards uniformly without replacement and ignore their order. What is the probability of *four of a kind*: four cards of the same rank, plus one other card? For example, $7\heartsuit, 7\spadesuit, 7\diamondsuit, 7\clubsuit, K\heartsuit$ qualifies.

There are $\binom{52}{5}$ equally likely hands. To construct a qualifying hand:

1. Choose the rank of the four matching cards: 13 choices.
2. Take all four cards of that rank: 1 choice.
3. Choose the fifth card from the remaining $52 - 4 = 48$ cards.

Every qualifying hand has exactly one four-card rank and one extra card, so this construction counts each hand once. Therefore

$$\Pr[\text{four of a kind}] = \frac{13 \cdot 1 \cdot 48}{\binom{52}{5}} = \frac{624}{2,598,960} \approx 0.0002401.$$

Unlike one prescribed hand, this event contains 624 different hands.

When draw order matters

For a question about the first or second draw, record an ordered sequence instead. Each r -element subset gives $r!$ different sequences:

$$|S_{\text{ordered}}| = \binom{N}{r} r! = \frac{N!}{(N-r)!} = N(N-1) \cdots (N-r+1).$$

These sequences are equally likely under uniform drawing without replacement. Either representation works for events that ignore order, provided numerator and denominator count the same kind of outcome.

Drawing without replacement is equivalent to taking the first r entries of a uniform random permutation. Each ordered sample has $(N-r)!$ completions. A full permutation is the case $r = N$; these models are related, not mutually exclusive categories.

Example: Drawing without replacement

Return to the ten labelled balls, but now keep each drawn ball outside the bag. What is the probability of drawing 3, 7, 1 in that order? There are $10 \cdot 9 \cdot 8 = 720$ equally likely ordered triples of distinct labels. Exactly one is the desired triple, so

$$\Pr[\text{draw 3, 7, 1 in order}] = \frac{1}{720}.$$

With replacement the answer was $1/1000$. The target is unchanged, but the sample space is smaller without replacement: sequences such as $(3, 3, 1)$ are no longer possible.

Practice: Red then blue. A bag contains three red and two blue balls. Draw two uniformly without replacement. What is the probability of red first, then blue? Distinguish the individual balls even when their colors match. There are $5 \cdot 4$ equally likely ordered draws and $3 \cdot 2$ favorable draws, so the probability is $6/20 = 3/10$.

Choosing the sample space

Use the experiment and whether order matters to choose the outcome representation. In the following models, each recorded outcome is equally likely.

Experiment	One outcome	Outcome count
r independent trials, each with m equally likely choices	An ordered sequence; repetition allowed	m^r
Shuffle N distinct objects uniformly	A complete ordering	$N!$
Draw r of N objects uniformly without replacement	An ordered sequence	$N!/(N-r)!$
The same draw, ignoring order	An r -element subset	$\binom{N}{r}$

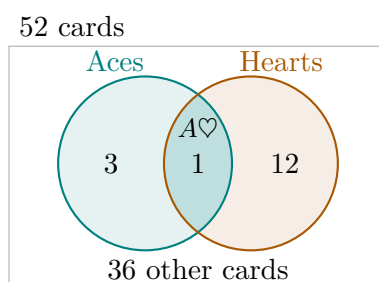
5 Event operations

MUST UNDERSTAND: Event operations

Let $A, B \subseteq S$ be events in the same sample space. An event is a set of outcomes, so event operations correspond to set operations:

Event operation	Corresponding set	When it occurs
NOT A	Complement: $\bar{A} = S \setminus A$	A does not occur.
A OR B	Union: $A \cup B$	At least one occurs, possibly both.
A AND B	Intersection: $A \cap B$	Both occur.

Running example: One card, two events. Draw one card uniformly from a standard 52-card deck, without jokers. Let A be the event *an ace* and B the event *a heart*. There are four aces and 13 hearts. The ace of hearts belongs to both: $A \cap B = \{A\heartsuit\}$.



NOT: Complement

The outcome weights in A and its complement sum to 1, so

$$\Pr[\overline{A}] = 1 - \Pr[A].$$

For our card draw, *not an ace* contains the other 48 cards:

$$\Pr[\text{not an ace}] = 1 - \frac{4}{52} = \frac{48}{52} = \frac{12}{13}.$$

OR: Union

Adding the probabilities of A and B counts their intersection twice. Subtract it once:

$$\Pr[A \cup B] = \Pr[A] + \Pr[B] - \Pr[A \cap B].$$

For an ace or a heart, the ace of hearts is counted twice. Therefore

$$\Pr[\text{ace or heart}] = \frac{4 + 13 - 1}{52} = \frac{16}{52} = \frac{4}{13}.$$

If the events are disjoint, the intersection term is zero.

AND: Intersection

For the same card draw, only the ace of hearts satisfies both conditions:

$$\Pr[A \cap B] = \frac{1}{52}.$$

AND means intersection: count the outcomes satisfying both conditions. Multiplying the two probabilities happens to give the same result here, but it is not a general rule for AND. L16 introduces the condition that permits multiplication.

Example: At least one ace

Now draw a five-card hand uniformly without replacement and ignore order. What is the probability of at least one ace? The outcomes are now hands, not individual cards, so there are $\binom{52}{5}$ equally likely outcomes. The complement is a hand with no aces: choose all five cards from the $52 - 4 = 48$ non-aces. Hence

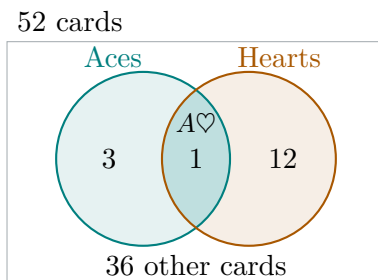
$$\Pr[\text{at least one ace}] = 1 - \frac{\binom{48}{5}}{\binom{52}{5}}.$$

Choosing a designated ace and four other cards would count hands with several aces more than once; the complement avoids this problem.

Example: At least one ace and one heart

Draw five cards uniformly without replacement and ignore order. What is the probability that the hand contains at least one ace and at least one heart? For example, $A\heartsuit, K\clubsuit, 2\diamondsuit, 7\spadesuit, 9\clubsuit$ qualifies.

The ace of hearts satisfies both requirements by itself. The hand fails if it contains no aces *or* no hearts. Let N_A be the event *no aces* and N_H the event *no hearts*. The desired event is the complement of $N_A \cup N_H$. There are 48 non-aces and 39 non-hearts. The ace of hearts belongs to both card groups, so the number of cards that are neither aces nor hearts is $52 - 4 - 13 + 1 = 36$:



The diagram counts individual cards; the events below consist of five-card hands. Thus

$$|N_A| = \binom{48}{5}, \quad |N_H| = \binom{39}{5}, \quad |N_A \cap N_H| = \binom{36}{5}.$$

Apply the complement rule and inclusion-exclusion:

$$\begin{aligned} \Pr[\text{at least one ace and one heart}] &= 1 - \Pr[N_A \cup N_H] \\ &= 1 - \Pr[N_A] - \Pr[N_H] + \Pr[N_A \cap N_H] \\ &= 1 - \frac{\binom{48}{5} + \binom{39}{5} - \binom{36}{5}}{\binom{52}{5}}. \end{aligned}$$

The intersection term adds back hands with neither aces nor hearts: these were subtracted twice.

Practice

In each problem, specify an equally likely sample space and the event before counting.

1. Roll two fair dice with all 36 ordered results equally likely. Find the probability of a sum of 9.
2. Generate a five-bit string uniformly. Find the probability of at least one 1.
3. Select one card uniformly from a standard deck. Find the probability of a spade or an ace.
4. Select one integer uniformly from $\{1, \dots, 100\}$. Find the probability it is divisible by 2 or 5.

Answers.

1. The pairs $(3, 6), (4, 5), (5, 4), (6, 3)$ give $4/36 = 1/9$.
2. The only excluded string is 00000, so the probability is $1 - 1/2^5 = 31/32$.
3. There are 13 spades, four aces, and one common card: $(13 + 4 - 1)/52 = 4/13$.
4. There are 50 multiples of 2, 20 of 5, and 10 of both: $(50 + 20 - 10)/100 = 3/5$.

Challenge exercise: Nondecreasing ranks

Draw five cards uniformly without replacement from a standard 52-card deck, keeping their order. What is the probability that their ranks are nondecreasing? Use the rank order

$$A < 2 < 3 < \dots < 10 < J < Q < K.$$

Equal ranks are allowed; suits do not affect the rank comparison. For example, $A\heartsuit, A\spadesuit, 5\diamondsuit, 6\diamondsuit, J\clubsuit$ is valid. The order $A\heartsuit, 5\diamondsuit, A\spadesuit, 6\diamondsuit, J\clubsuit$ is invalid because the rank drops from 5 to A.

Solution. Let E be the event that the ranks are nondecreasing. All ordered five-card draws are equally likely, so

$$|S| = 52 \cdot 51 \cdot 50 \cdot 49 \cdot 48 = 311,875,200.$$

Count E by grouping the cards according to repeated ranks. The distinct rank groups must appear in increasing order, but cards of the same rank may appear in any suit order. Thus, sorting the ranks does *not* always leave just one valid order for a given hand: $1/5!$ is not the answer.

For example, the pattern $2 + 1 + 1 + 1$ means one pair and three other distinct ranks. Choose the pair's rank in 13 ways and the other ranks in $\binom{12}{3}$ ways. Their rank order is fixed. Within the pair, choose the first suit in 4 ways and the second in 3 ways; each remaining card has 4 suit choices. This gives

$$13 \binom{12}{3} (4 \cdot 3) 4^3 = 2,196,480.$$

For any group of j cards with the same rank, the ordered suit choices are $4 \cdot 3 \cdots (4 - j + 1)$. Apply this rule to each group:

Rank pattern	Choose ranks	Suit orders	Sequences
$1 + 1 + 1 + 1 + 1$	$\binom{13}{5}$	4^5	1,317,888
$2 + 1 + 1 + 1$	$13 \binom{12}{3}$	$(4 \cdot 3) 4^3$	2,196,480
$2 + 2 + 1$	$\binom{13}{2} \cdot 11$	$(4 \cdot 3)^2 \cdot 4$	494,208
$3 + 1 + 1$	$13 \binom{12}{2}$	$(4 \cdot 3 \cdot 2) 4^2$	329,472
$3 + 2$	$13 \cdot 12$	$(4 \cdot 3 \cdot 2)(4 \cdot 3)$	44,928
$4 + 1$	$13 \cdot 12$	$(4 \cdot 3 \cdot 2 \cdot 1) 4$	14,976

These cases are disjoint and cover every possible rank pattern: five equal ranks are impossible because each rank has only four cards. Adding the six counts gives $|E| = 4,397,952$. Therefore

$$\Pr[E] = \frac{4,397,952}{311,875,200} = \frac{881}{62,475} \approx 1.4102\%.$$

Summary

Concept	Definition or rule
Sample space and event	S : all outcomes; $E \subseteq S$: the desired outcomes.
Classical probability	$\Pr[E] = E / S $, for finite nonempty uniform S .
General discrete model	$\Pr[E] = \sum_{x \in E} \Pr[x]$.
Reduced outcomes	Equal full-outcome counts preserve uniformity.
NOT	$\Pr[\overline{E}] = 1 - \Pr[E]$.
OR	$\Pr[A \cup B] = \Pr[A] + \Pr[B] - \Pr[A \cap B]$.
AND	The event $A \cap B$.

Optional: Bertrand's paradox

Choose a random chord of a circle. Is it longer than a side of the inscribed equilateral triangle? Our finite counting formula does not apply to infinitely many chords. Choosing two independent uniform endpoints on the circumference gives probability $1/3$; choosing the midpoint uniformly by area inside the disk gives $1/4$. Different procedures define different experiments: “random” alone is insufficient. These geometric results are not derived here.